Jewel Nath

Software Engineer · AI Agents & Automation · Robotics

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SUMMARY

Software engineer since 2024, focused on AI. At Orbitax, I build .NET backend services and Angular features for a tax technology platform for multinational companies. At RoboAI, the startup I founded, I built AI products, such as a team workspace where AI assistants do real work and an electronics shop whose AI plans the parts for a student's robot. At Orbitax, I also build OrbiBot in my free time: its first version mapped a room, drove itself and took commands from an AI agent.

EXPERIENCE

Orbitax · Software Engineer

Aug 2026 – present · Associate Software Engineer, May 2024 – Jul 2026

A tax technology platform for multinational companies.

- Built features that help tax teams track and file every deadline across countries, including the OECD global minimum tax (Pillar Two): yearly roll-forward, duplicate checks, linked forms and a deadline dashboard.
- Led the reliability rebuild of a core platform feature, private workspaces where a company works on a copy of its data, from design decisions to documentation and release.
- Own production issues in my areas from first report to verified fix.

Stack: C#, .NET, ASP.NET Core, background workflows, MongoDB, SQL, Angular, TypeScript, NgRx, Node.js, Python, NUnit, pytest

RoboAI · Technical Advisor

Aug 2026 – present · Robotics & AI Automation Engineer, part time, May 2024 – Jul 2026 · Founder & CEO, Oct 2023 – Apr 2024 · Dhaka, Bangladesh

A startup I founded that builds AI and robotics products for teams, shops and learners. It began in 2023 by teaching robotics and programming to students, and relaunched as an AI studio in 2026. My juniors now run it, and I advise them.

- As founder and CEO: led the team's robotics R&D, taught robotics and programming, and guided students' projects for robot contests and showcases.
- Built **AI powered Team OS** (teamos.site), one app where a small company runs its projects, people, time and expenses, and can ask AI to do the work. ChatGPT and Claude connect to it through an MCP (Model Context Protocol) server, and every AI action runs with the user's own permissions; an AI can request a payment, but only a person can approve it.
- Built the AI parts planner for **RoboAI Shop** (shop.roboai.one): a student describes a robot project and the AI picks the parts from stock. Prices and totals come only from the database, and the server strips any part or price the AI invents.
- Built robotics projects with the team, including a fire-fighting robot, line followers, a robotic arm and a conveyor-belt sorting system, and help school, college and university students design and build their own robots.

Stack: AI (MCP, OAuth 2.1, Gemini, ChatGPT and Claude connectors), web (TypeScript, Next.js, React, Supabase with Postgres, Vercel), robotics (Arduino, ESP32, Raspberry Pi, ROS 2, OpenCV, sensors and motor control)

School of Robotics, Cumilla · Programming & Robotics Trainer

Mar 2021 – Mar 2024 · part time

- Taught school and college students to program and build robots, in workshops with the Cumilla Deputy Commissioner's Office across 10 upazilas and 30 schools and colleges, and coached students for the National High School Programming Contest.

Quanta Robotics · Co-Founder & Robotics Programmer

May 2019 – Dec 2023

A robotics team I co-founded at university, which built humanoid robots.

- Worked on every part of its main robot, Robot Niko: the 3D printing, the assembly, the servo tuning, the arm motions and the software (see Projects).
- Niko was launched at the Deputy Commissioner's Office in December 2022 and covered by the national press, among them Kaler Kantho, The Daily Post and BTV.

PROJECTS

- **OrbiBot** (2025 – present, at Orbitax, in my free time). Its first version, V1, was a mecanum-wheel robot I built myself, from the welded steel frame and electronics to the software. Its third version, V3, Orbitax's office telepresence robot, is designed in Python-scripted CAD and tested in MuJoCo simulation.
  - Mapped a room and navigated on its own: LiDAR SLAM, Nav2 path planning, and an EKF that fuses wheel and IMU data, on ROS 2.
  - Took commands from an AI agent with 58 robot tools on Gemini, Claude or Ollama, also over an MCP server.
  - Ran on my own STM32 motor firmware with PID control and a safety stop, with a React dashboard for driving and live video (WebRTC).
- **Robot Niko** (2022). A life-size, 3D-printed humanoid assistant in the InMoov design, built with the team at Quanta Robotics, on a Raspberry Pi.
  - 3D-printed and assembled its body, tuned its 27 servos on two PCA9685 boards, and planned its arm motions as servo sequences.
  - Led its programming and wrote most of its Python code: a "Hey Niko" wake word, spoken conversation, and face recognition that greets each person by name with a handshake, while its head turns to follow a face (OpenCV).
- **EnviroBot** (2023 – 2024). Built at Comilla University with my team: a rover with a robot arm that tells paper, plastic and other waste apart with YOLOv8, and picks it up, using Arduino control and a ROS 2 model of the arm (URDF, RViz).
- **Sign Language Detection** (2023). Reads American Sign Language hand signs from a camera and turns them into letters, using MediaPipe hand tracking and a PyTorch CNN; written up as a team project report.

SKILLS

Languages:	C#, Python, TypeScript, JavaScript, C++
AI engineering:	AI agents, MCP servers, LLM tool calling, RAG, structured outputs, Ollama, LiteLLM
AI dev tools:	Claude Code (skills, subagents, rules), Codex, Cursor, AI code review, spec-driven development
Backend & data:	.NET 10, ASP.NET Core, CQRS, background workflows, Node.js, MongoDB, PostgreSQL, Supabase
Frontend:	Angular, Next.js, Tailwind
Testing:	Playwright, NUnit, pytest, Vitest
DevOps & infra:	Git, Linux, Docker, Vercel, Cloudflare R2, systemd services on Raspberry Pi
Robotics & vision:	ROS 2 Jazzy, Nav2, SLAM, robot_localization, MoveIt, URDF, RViz, Gazebo, MuJoCo; OpenCV, YOLOv8, MediaPipe, PyTorch
Embedded & hardware:	STM32 firmware, PID control, ESP32, Arduino, Raspberry Pi, KiCad, FreeCAD (Python-scripted), 3D printing
Spoken languages:	Bangla (native), English (professional)

EDUCATION

Comilla University · B.Sc. Engineering, Information and Communication Technology

2019 – 2024 · CGPA 3.43 / 4

ACHIEVEMENTS

- Top 100 of 1,051 ideas, Mujib 100 Idea Contest 2021 (University Grants Commission and Ministry of Education, part of IC4IR 2021), for our team's waste-management plan for Cumilla City.
- Runner-up, Line Follower Robot contest, BAIUST EEE Day 2023 (team CoU Bots).
- Codeforces Specialist (max rating 1509), 1,200+ problems solved (handle CodeCell).